

Learning Geometric Cuts for Nearest Neighbor Search

Sanjeev Khanna

NYU

Ashwin Padaki

UPenn

Erik Waingarten

UPenn

What I hope you take away

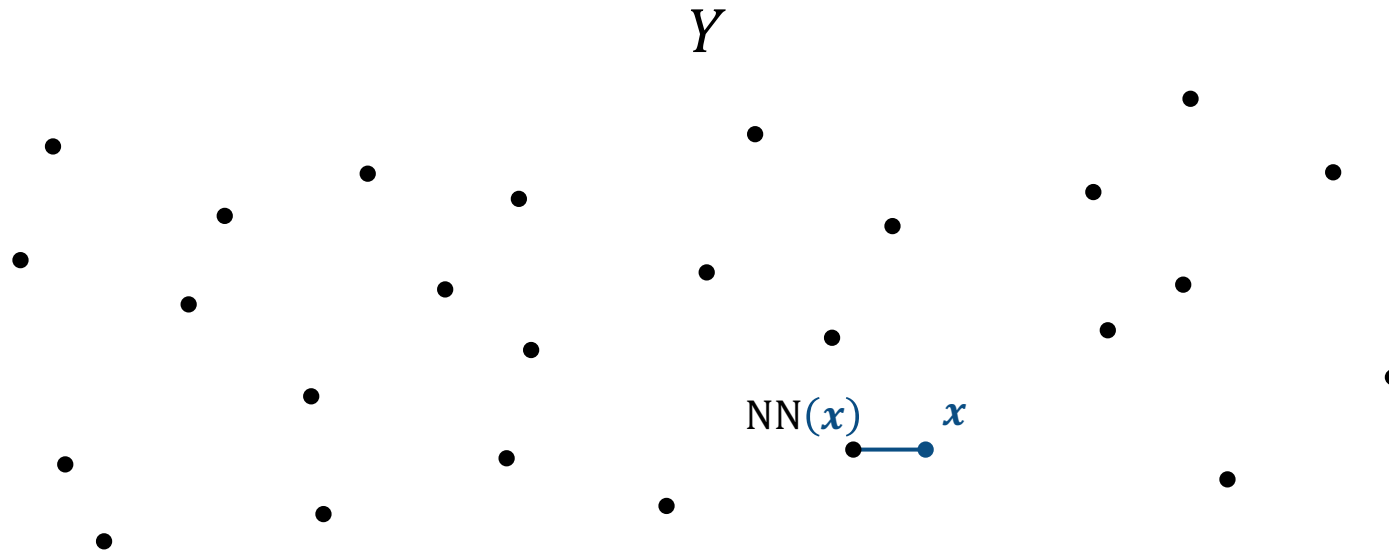
1. **New perspective:** nearest neighbors beyond the worst case
2. **New problem:** finding the best geometric cut
3. **Cool techniques:** random thresholding + polynomial approximation

Nearest Neighbor Search

Dataset Y , query x

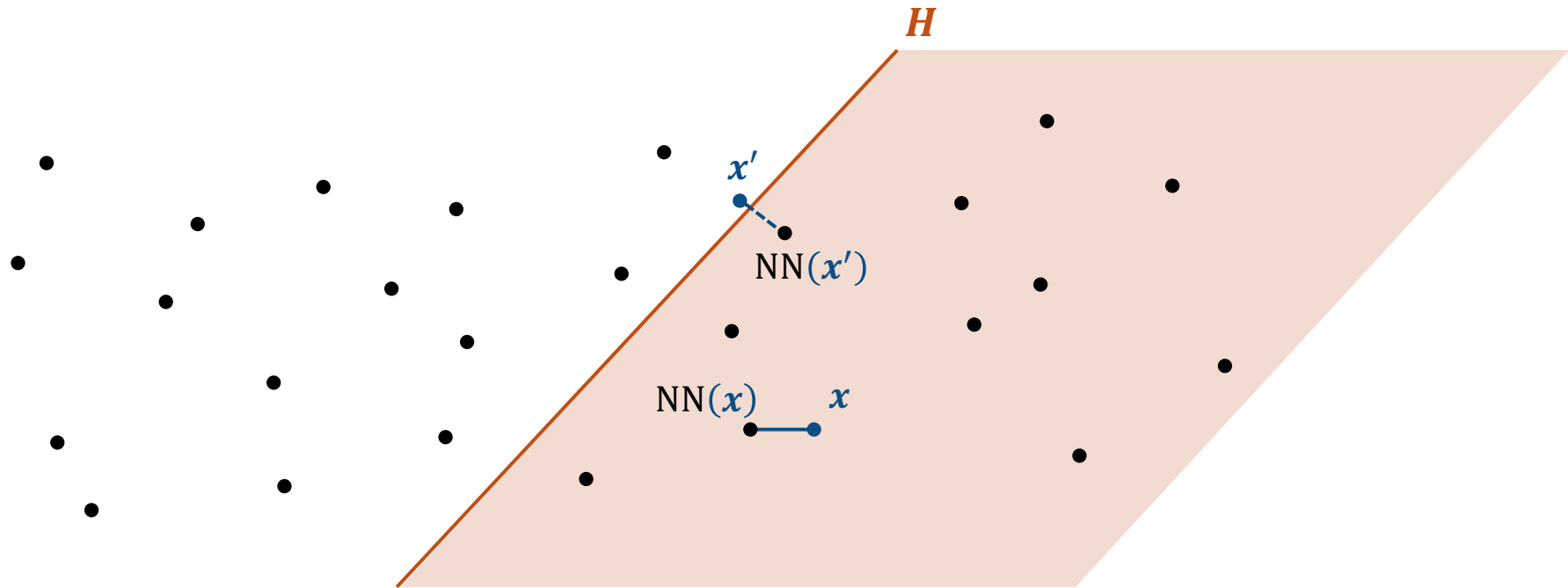
Task: find $\text{NN}(x) = \arg \min \|y - x\|$

KD-Tree,
LSH,
HNSW,
DiskANN,
IVF-PQ,
....



“Cut the Search Space”

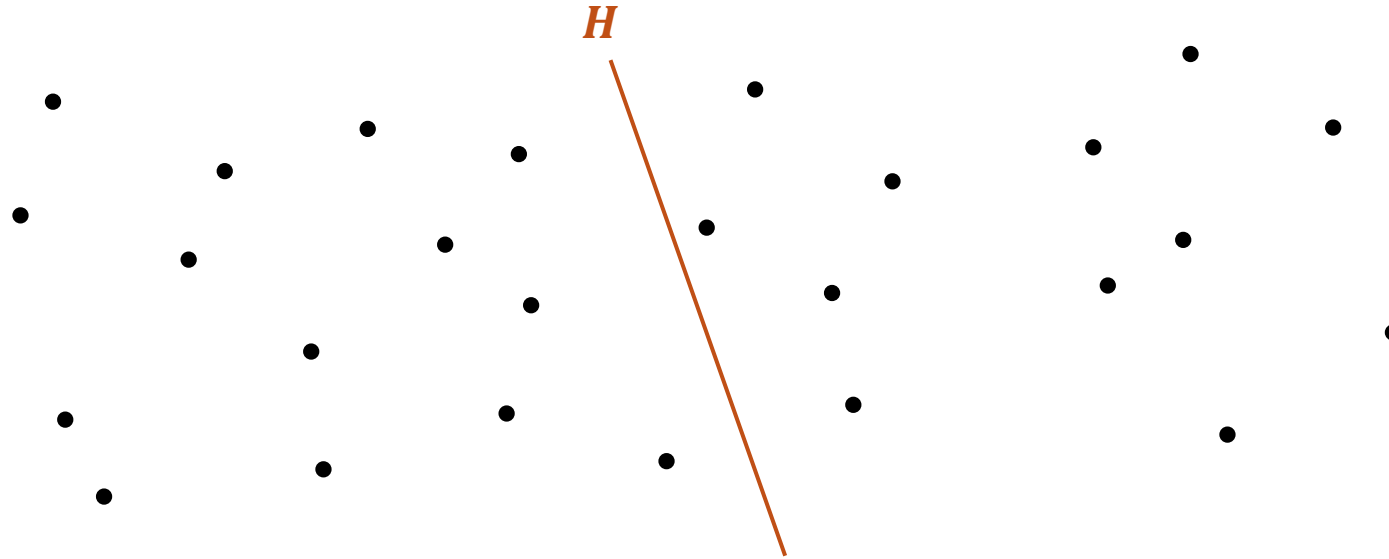
Simplest cut: hyperplane!



Q. How to choose hyperplane?

Approach 1: *prepare for the worst*

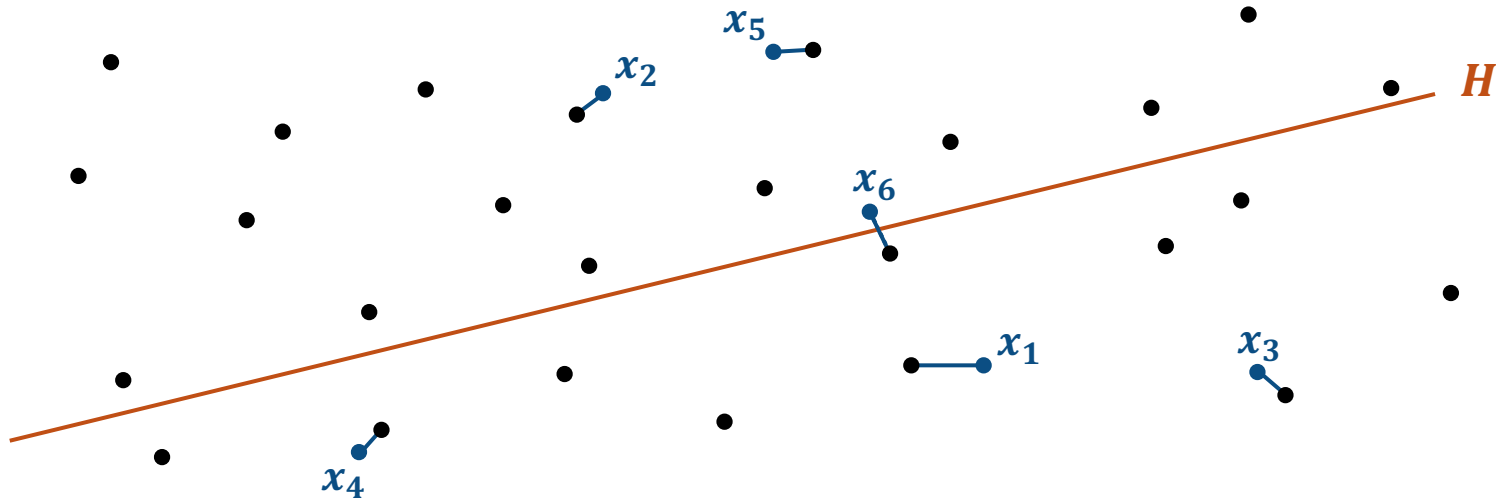
We don't know x , so choose H randomly!



“Random projection LSH”

Our Approach: *learn from the past*

Use observations $x_1, x_2, x_3, \dots$ to choose H

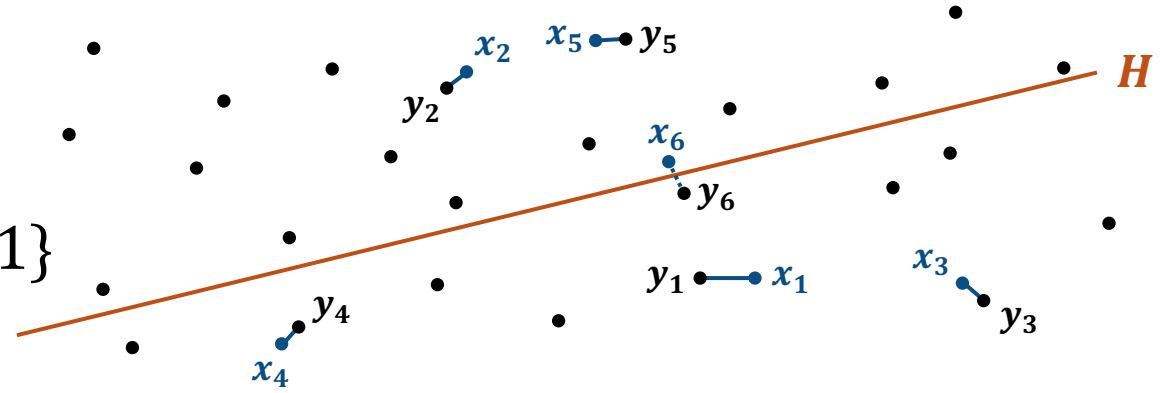


$$\text{err}(H) = 1/6$$

Best Hyperplane Cut

Input: samples $(\mathbf{x}_i, \mathbf{y}_i) \sim \mathcal{D}$,

Output: hyperplane cut $\mathbf{H} : \mathbb{R}^d \rightarrow \{-1, 1\}$



1. Preserves neighbors

$$\text{minimizes } \text{err}_{\mathcal{D}}(\mathbf{H}) := \Pr_{\mathcal{D}}[\mathbf{H}(\mathbf{x}) \neq \mathbf{H}(\mathbf{y})]$$

2. Balances the dataset

$$\text{bal}_{\mathcal{D}}(\mathbf{H}) := \min_{b \in \{-1, 1\}} \Pr_{\mathcal{D}}[\mathbf{H}(\mathbf{y}) = b] \geq 1/3$$

Three Main Results

Given samples $(\mathbf{x}_i, \mathbf{y}_i) \sim \mathcal{D}$, find a hyperplane cut H that:

minimizes $\text{err}_{\mathcal{D}}(H)$

subject to $\text{bal}_{\mathcal{D}}(H) \in [1/3, 2/3]$

1. NP-hardness result (one slide)
2. **Improper learning algorithm** (**main focus**)
3. Guarantees for nearest neighbor search (one slide)

Result 1: An **NP-Hard** Problem

Input: pairs $(x_1, y_1), \dots, (x_m, y_m) \in \mathbb{R}^d$

Promise: exists H^* with: $\text{err}(H^*) = 0, \quad \text{bal}(H^*) = 1/2$

Task: find any H with: $\text{err}(H) \leq 0.01, \quad \text{bal}(H) \geq 1/10$

Rules out “ $f(\text{OPT}) + \varepsilon$ ” error guarantee

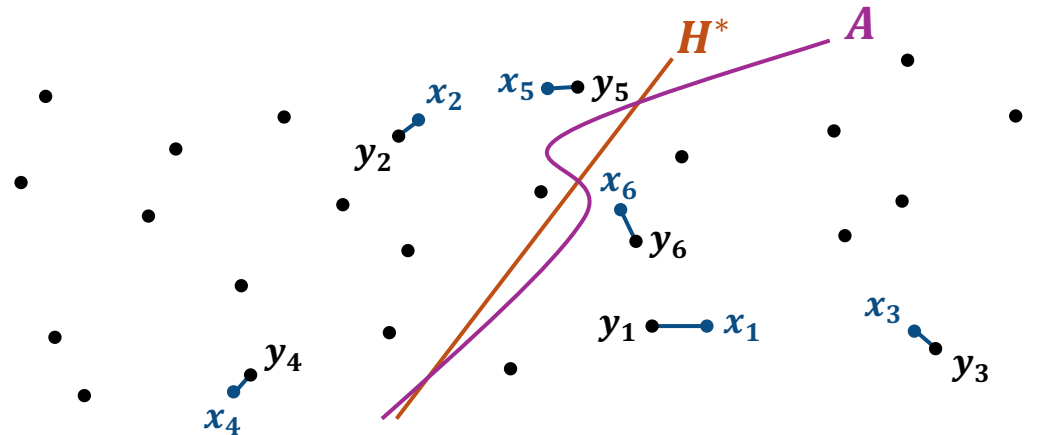
Result 2: An Improper Learning **Algorithm**

Assumption: $\mathcal{D}$ has Gaussian marginals

$$H^* = \min_H \{ \text{err}(H) \mid \text{bal}(H) \geq 1/3 \}$$

Guarantee: Output degree- $(1/\varepsilon^2)$ polynomial cut $A : \mathbb{R}^d \rightarrow \{-1, 1\}$,

$$\text{err}(A) \leq \sqrt{\text{err}(H^*) + \varepsilon} \quad \text{and} \quad \text{bal}(A) \geq 1/10$$



d^{1/ε^2} runtime and samples

Two Techniques

Assumption: $\mathcal{D}$ has Gaussian marginals

Guarantee: Output degree- $(1/\varepsilon^2)$ polynomial cut $\mathbf{A} : \mathbb{R}^d \rightarrow \{-1, 1\}$,

$$\text{err}(\mathbf{A}) \leq \sqrt{\text{err}(\mathbf{H}^*)} + \varepsilon \quad \text{and} \quad \text{bal}(\mathbf{A}) \geq 1/10$$

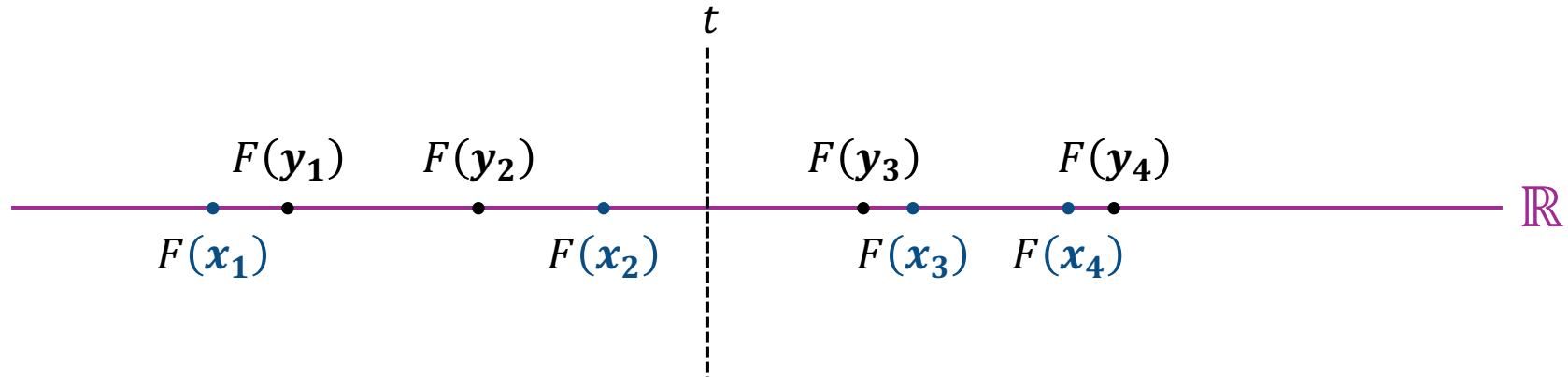
1. **Random thresholding** (“Cheeger’s inequality”)
2. **Polynomial approximation** (“PAC-learning halfspaces”)

Cut = Projection + Threshold

A geometric cut has two components

(1) **Projection** $F : \mathbb{R}^d \rightarrow \mathbb{R}$

(2) **Threshold** $t \in \mathbb{R}$

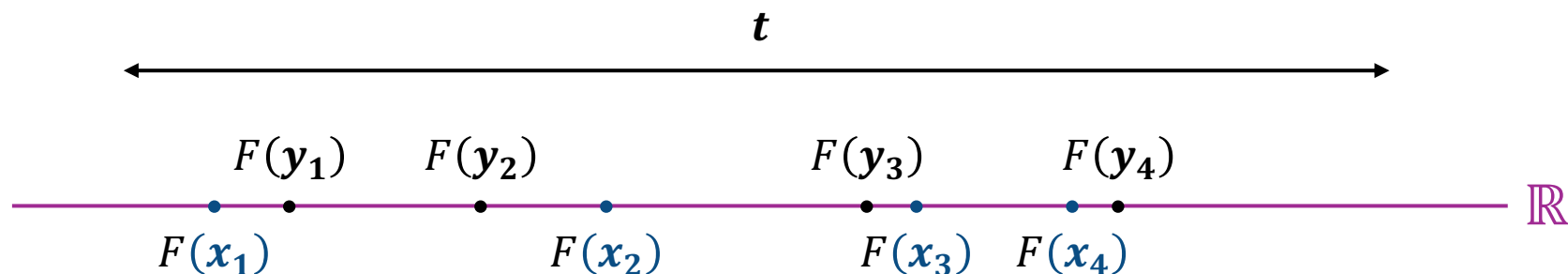


Random thresholding: optimize over F assuming **random** t

Random Thresholding

Projection $F : \mathbb{R}^d \rightarrow \mathbb{R}$

Random threshold $t \in \mathbb{R}$



$$\Pr_t[(x, y) \text{ is cut}] \propto |F(x) - F(y)|$$

$$\mathbb{E}_{\mathcal{D}, t}[\text{err}(F, t)] \propto \mathbb{E}_{\mathcal{D}}[|F(\mathbf{x}) - F(\mathbf{y})|]$$

A Proxy for Random Threshold Error

Projection $F : \mathbb{R}^d \rightarrow \mathbb{R}$

Random threshold $t \in \mathbb{R}$

$$\mathbb{E}_{\mathcal{D}, t}[\text{err}(F, t)] \propto \mathbb{E}_{\mathcal{D}}[|F(\mathbf{x}) - F(\mathbf{y})|]$$

$$\mathbb{E}_{\mathcal{D}, t}[\text{bal}(F, t)] \propto \mathbb{E}_{\mathcal{D}}[|F(\mathbf{y}_1) - F(\mathbf{y}_2)|]$$

Proxy:

$$\mathcal{R}(F) := \frac{\mathbb{E}[|F(\mathbf{x}) - F(\mathbf{y})|]}{\mathbb{E}[|F(\mathbf{y}_1) - F(\mathbf{y}_2)|]}$$

Low Proxy implies Low Error

$$\mathcal{R}(F) := \frac{\mathbb{E}[|F(\mathbf{x}) - F(\mathbf{y})|]}{\mathbb{E}[|F(\mathbf{y}_1) - F(\mathbf{y}_2)|]}$$

Averaging argument

for any F , there exists t with

$$\text{err}(F, t) \leq \mathcal{R}(F)$$

The algorithm

1. Minimize $\mathcal{R}(F)$ over linear functions....

Nonconvex :(

Low Proxy implies Low Error

$$\mathcal{R}(F) := \frac{\mathbb{E}[|F(\mathbf{x}) - F(\mathbf{y})|^2]}{\text{Var}[F(\mathbf{y})]}$$

Averaging argument

for any F , there exists t with

$$\text{err}(F, t) \leq \sqrt{\mathcal{R}(F)}$$

The algorithm

1. Minimize $\mathcal{R}(F)$ over linear functions or deg- k polys
[bottom eigvec of $d^k \times d^k$ matrix]
2. Choose the best threshold t

Caveat: no balance guarantee,
so we use SDP relaxation

Low Error implies Low Proxy?

Algorithm: finds cut of error $\leq \sqrt{\min_F \mathcal{R}(F)}$

The other direction: consider best hyperplane cut $\mathbf{H}^* = (h^*, t^*)$

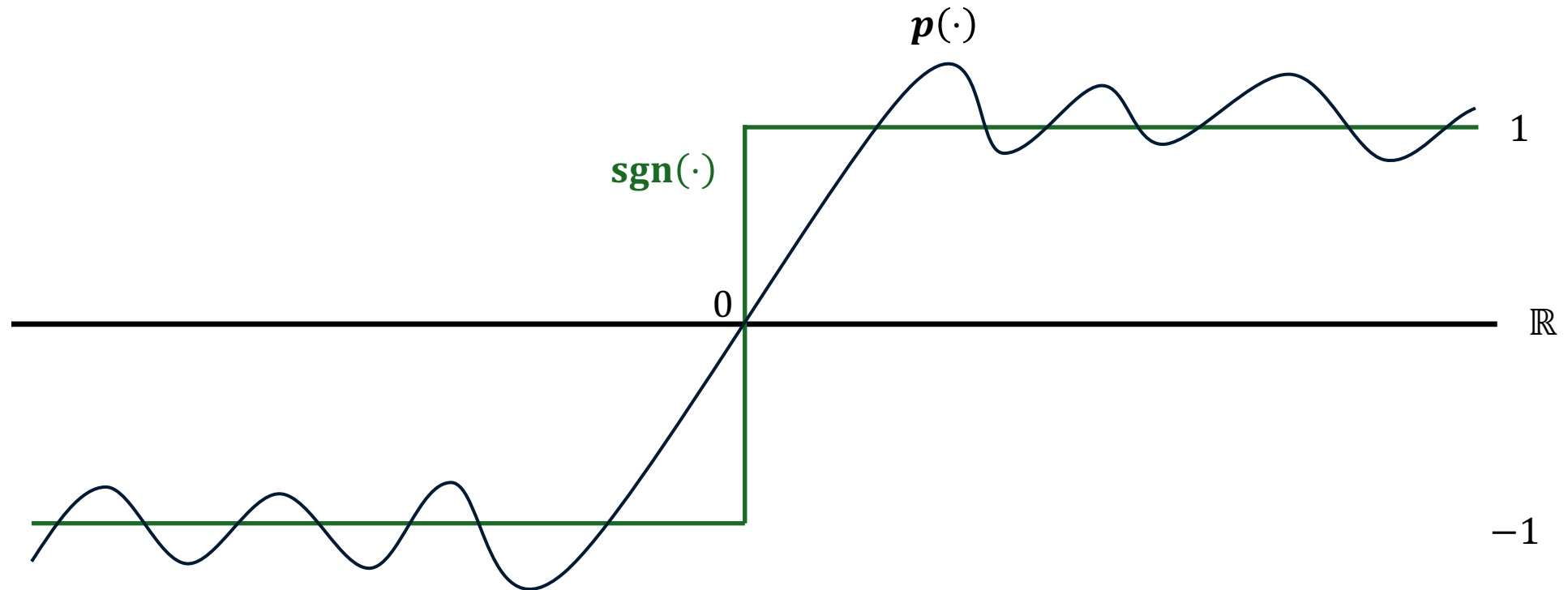
$$\mathcal{R}(\mathbf{H}^*) = \frac{\mathbb{E}_{\mathcal{D}}[|\mathbf{H}^*(\mathbf{x}) - \mathbf{H}^*(\mathbf{y})|^2]}{\text{Var}[\mathbf{H}^*(\mathbf{y})]} = \Theta(\text{err}_{\mathcal{D}}(\mathbf{H}^*))$$

Issue: $\mathbf{H}^*(z) = \text{sgn}(h^*(z) - t^*)$ is not linear

The Fix: Polynomial Approximation

Issue: $H^*(z) = \text{sgn}(h^*(z) - t)$ is not linear

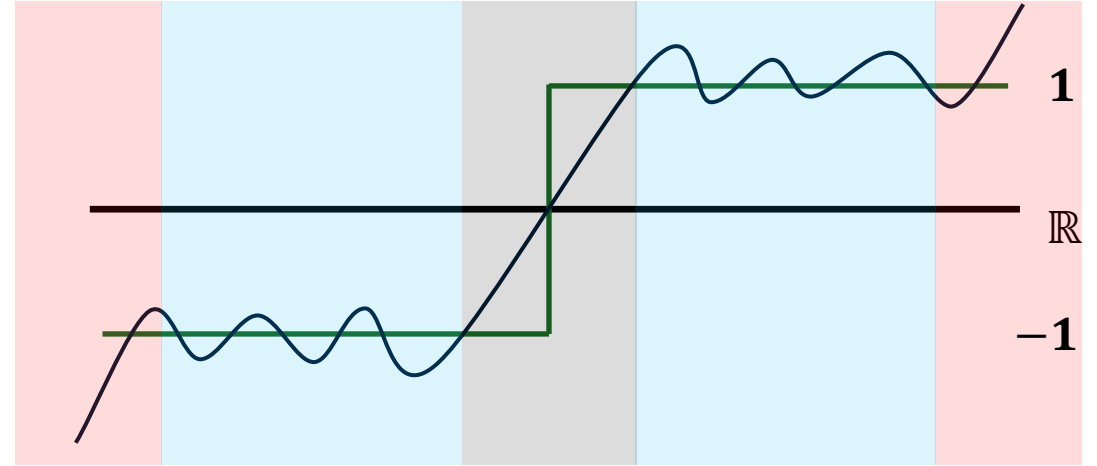
Solution: approximate $\text{sgn}(\cdot)$ with a low-degree polynomial!



Polynomial Approximation of Sign Function

Three regions:

- **Interval:** $|p(z) - \text{sgn}(z)| \leq \varepsilon$
- **Transition:** $|p(z) - \text{sgn}(z)| \leq 1$
- **Tail:** $p(z) \approx |z|^k$



To limit error of transition and tail:

1. Exponentially decaying mass on **Tail**
2. Only small amount of mass on **Transition**

What distributions $\mathcal{D}$ are nice?

For **both** marginals of $\mathcal{D}$, **every** linear projection h places:

1. Exponentially decaying mass on **Tail**
2. Only small amount of mass on **Transition**

$\mathbf{x}$ - and $\mathbf{y}$ -marginal are Gaussian $\rightarrow$ error ε with degree $1/\varepsilon^2$

In general: above tradeoff holds if $\mathbf{x}$ - and $\mathbf{y}$ -marginals satisfy:

1. Subgaussian tails: $\Pr[\langle h, \mathbf{x} \rangle \geq t] \leq \exp(-\Omega(t^2))$
2. Anticoncentration: $\Pr[|\langle h, \mathbf{x} \rangle| \leq \ell] \leq O(\ell)$

The Result

Assumption: $\mathcal{D}$ has Gaussian-like marginals

Guarantee: Output degree- $(1/\varepsilon^2)$ polynomial cut $\mathbf{A} : \mathbb{R}^d \rightarrow \{-1, 1\}$,

$$\text{err}(\mathbf{A}) \leq \sqrt{\text{err}(\mathbf{H}^*) + \varepsilon} \quad \text{and} \quad \text{bal}(\mathbf{A}) \geq 1/10$$

Time / Sample Complexity: $d^{O(1/\varepsilon^2)}$

Open Question: is it possible to achieve time $\text{poly}(d, 1/\varepsilon)$?

Guarantees for Nearest Neighbor Search

- **Idea:** run our single-cut algorithm recursively to obtain a tree
- Balancedness $\Rightarrow$ #dataset points decays with depth
- Low error $\Rightarrow$ most queries stay with their NN

Final Theorem

If $\mathcal{D}$ has Gaussian-like marginals + admits a perfect halfspace tree, we obtain a tree of polynomial cuts achieving $o(nd)$ query time.

What I hope you took away

1. **New perspective:** nearest neighbors beyond the worst case
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Thanks!